

Figure 1: Mobile hybrid application

characteristics for mobile hybrid apps:

- UI implementation is done using Web technologies (HTML, CSS and JavaScript)
- Apps are capable of using the mobile platform or operating system APIs
- An executable file installs and resides at the mobile device
- The app can be distributed via a platform-specific app store or an enterprise distribution mechanism

Figure 1 depicts a representation of a mobile hybrid application. Broadly, it has two layers - the container and the user interface (UI), where the container is implemented on native mobile technology (Android, iOS or Windows Phone) and the UI is implemented with Web technologies (HTML, CSS and JS).

There are many frameworks and tools for mobile hybrid app development. The major selling point of mobile hybrid apps is cross-platform development. The following are the features that can be leveraged with mobile hybrid apps:

- Integration of open source frameworks with HTML5
- Liquid layouts for multi-screen UIs
- Local storage, multimedia handling, semantics and forms, graphics, etc
- A single code-based architecture model for multi-platform presentations
- Mobile hybrid framework that bundles the HTML5 based view layer with native platform containers to create deployable builds

The mobile hybrid application development space has witnessed increasing interest from different sets of customers and companies. Over the last few years, many frameworks have come up, which include a mix of open source and licensed frameworks for mobile hybrid applications. Most of them cover particular aspects of mobile hybrid app development.

appEZ

appEZ, an open source multi-architecture/multi-platform

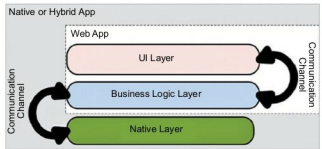


Figure 2: Three distinctive layers of appEZ

mobile app development framework, has a unique flexible architecture for true mobile hybrid app development with flawless amalgamation of native (Android/iOS/WP) and Web (HTML5, CSS and JavaScript) technologies, depending on the app's need. It covers all the layers of mobile hybrid application development. Its modular architecture makes it possible to use only the required components, based on the business requirements as against using the complete library set-up.

appEZ: The application layer structure

The appEZ mobile application incorporates multiple layers in one application code base.

Figure 2 illustrates three distinctive layers. These are listed below.

- **UI layer:** This represents the UI/UX and view implementation for the appEZ mobile app. The appEZ platform has multiple ways of implementing the UI layer as hybrid/Web or native (Android Activity). The hybrid and Web application UI layer is implemented with HTML, CSS, JS and other UI frameworks/tools (jQuery, jQueryMobile, LESS, Bootstrap, etc). This layer supports UI/UX guidelines for app-centric design or platform-specific adaptive design, depending on the application's needs.
- **Business logic layer:** This layer is responsible for the core logic and handles the implementation for the application. It also communicates with the server for the required data and information. The appEZ-recommended approach and respective design patterns can be used to implement the application's features. In case of cross-platform development, it is recommended that users harness the common business logic layer among platform-specific builds.
- **Native layer:** This layer takes care of the platform-specific capabilities (camera, database, HTTP communication, persistent store, etc) to be used by appEZ. This layer is not accessible with Web application implementation. This is the reason Web applications possess limited access to mobile platform capabilities.
Mobile applications powered by appEZ comprise three components, as illustrated in Figure 3.